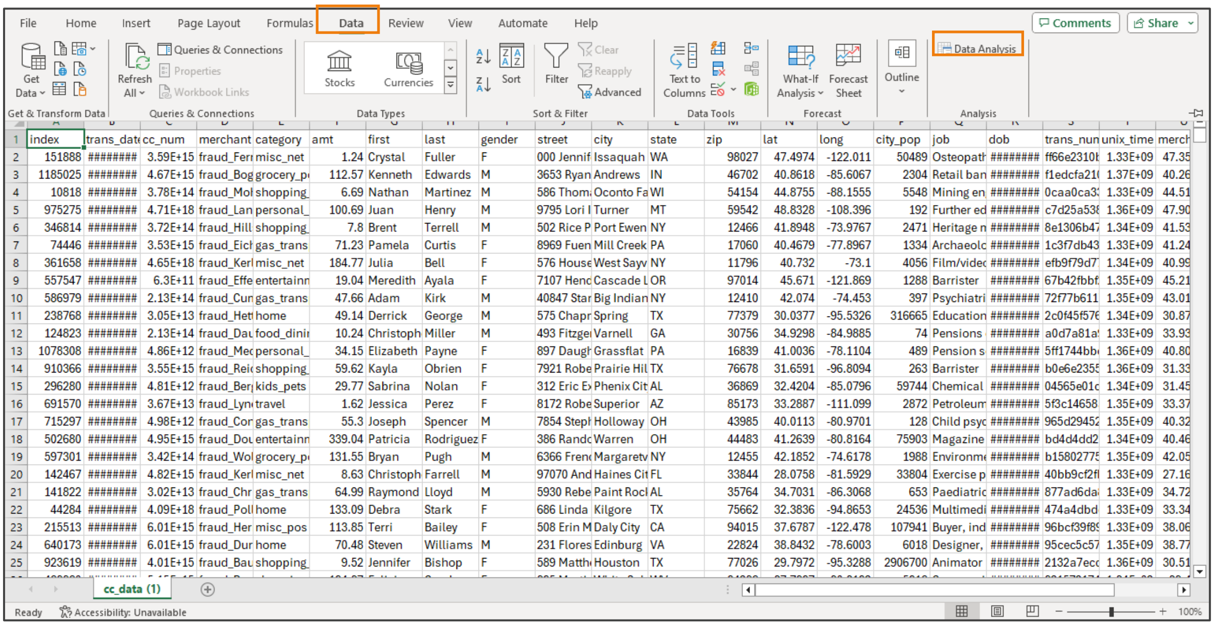
Capstone 1: Excel

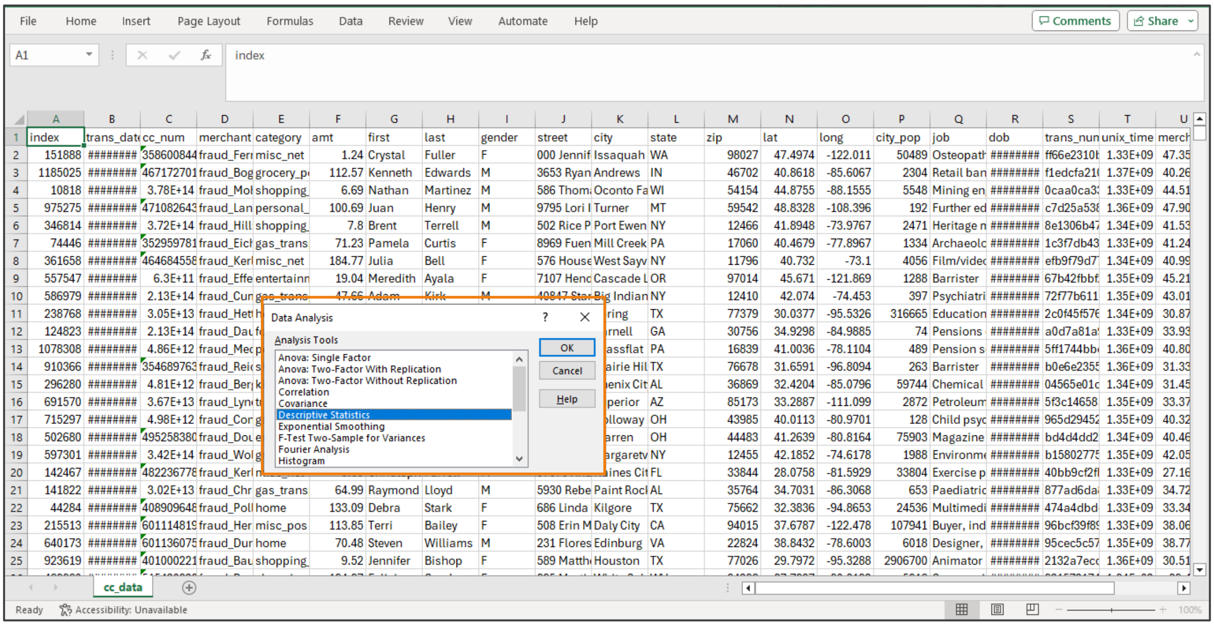
Data Exploration

1. Create statistical summary on amt and city_pop

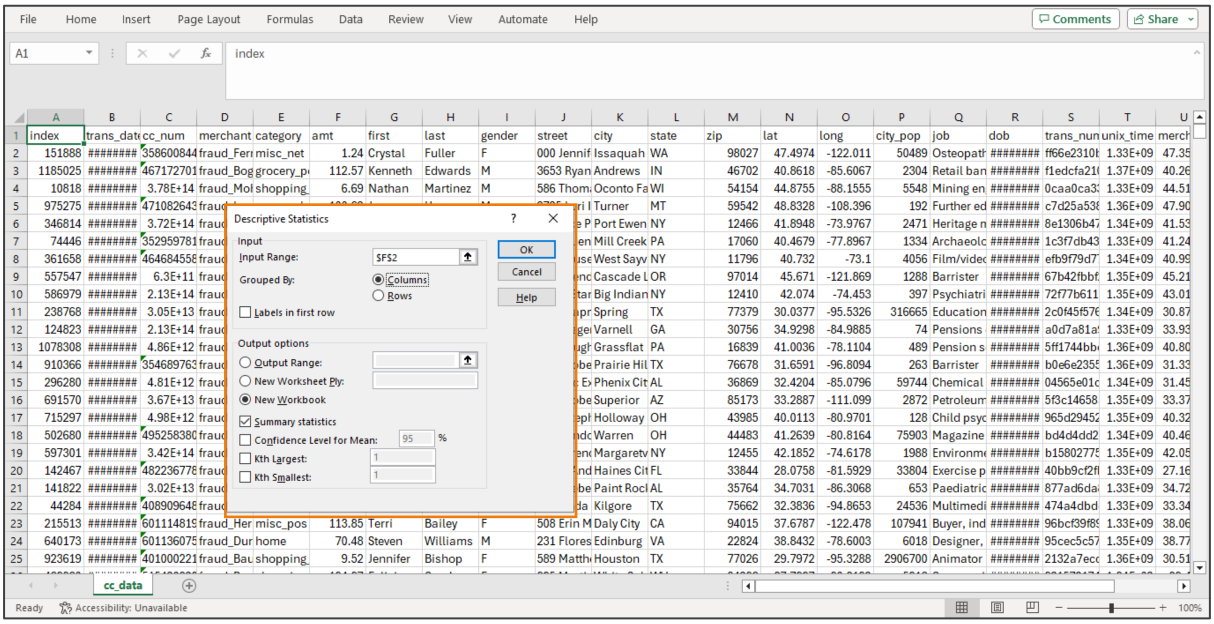
1.1 Go to the **Data** tab and click on **Data Analysis** (If it is not visible, you might need to enable it in Excel options.)



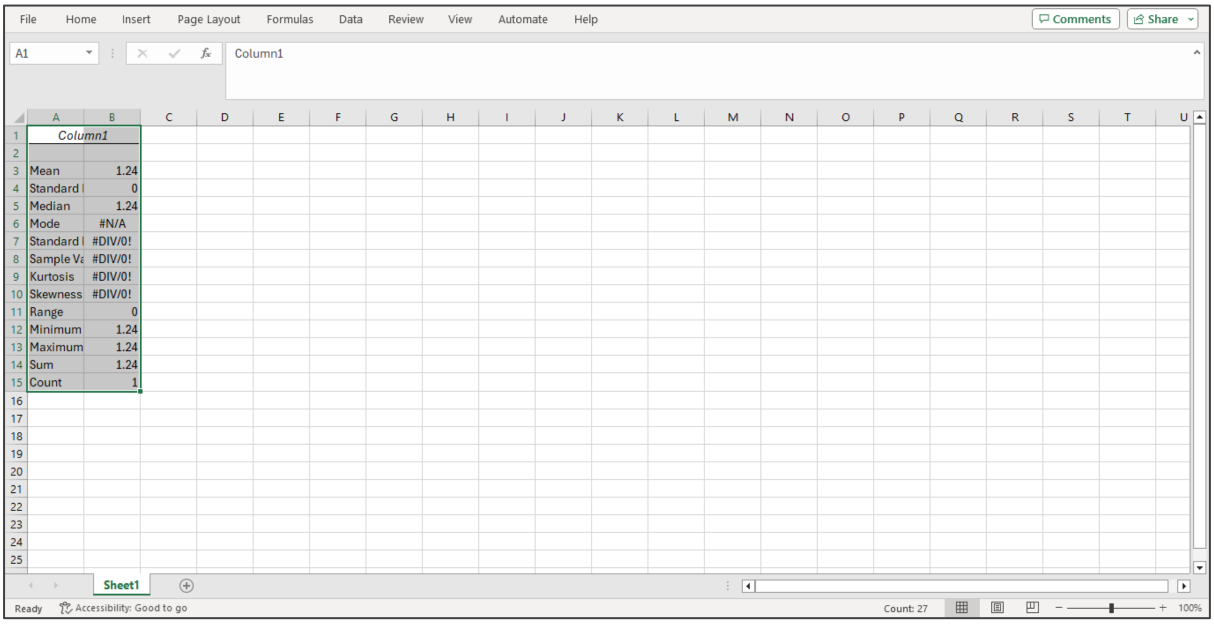
1.2 Select **Descriptive Statistics** from the list and click **OK**



1.3 Input the range for **amt**, check the **Summary statistics** option, select an output option as **New Workbook**, and click **OK**

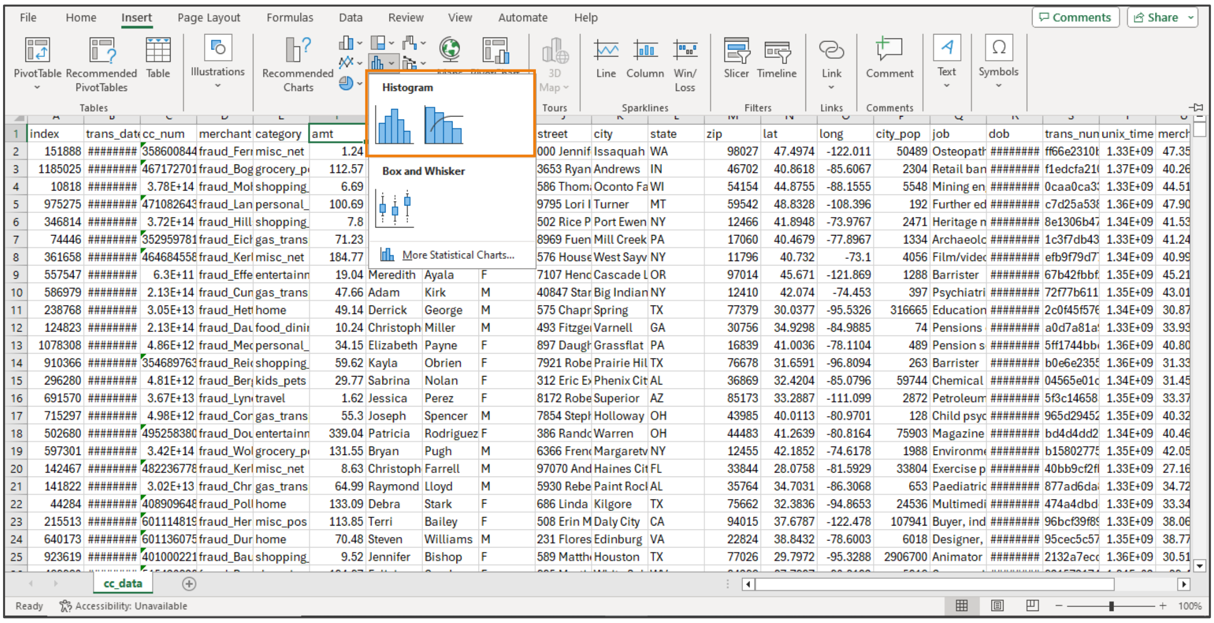


Here is the sample output for input range of column **amt** with data cell F2, which creates a new notebook. Same can be done for **city_pop** column.

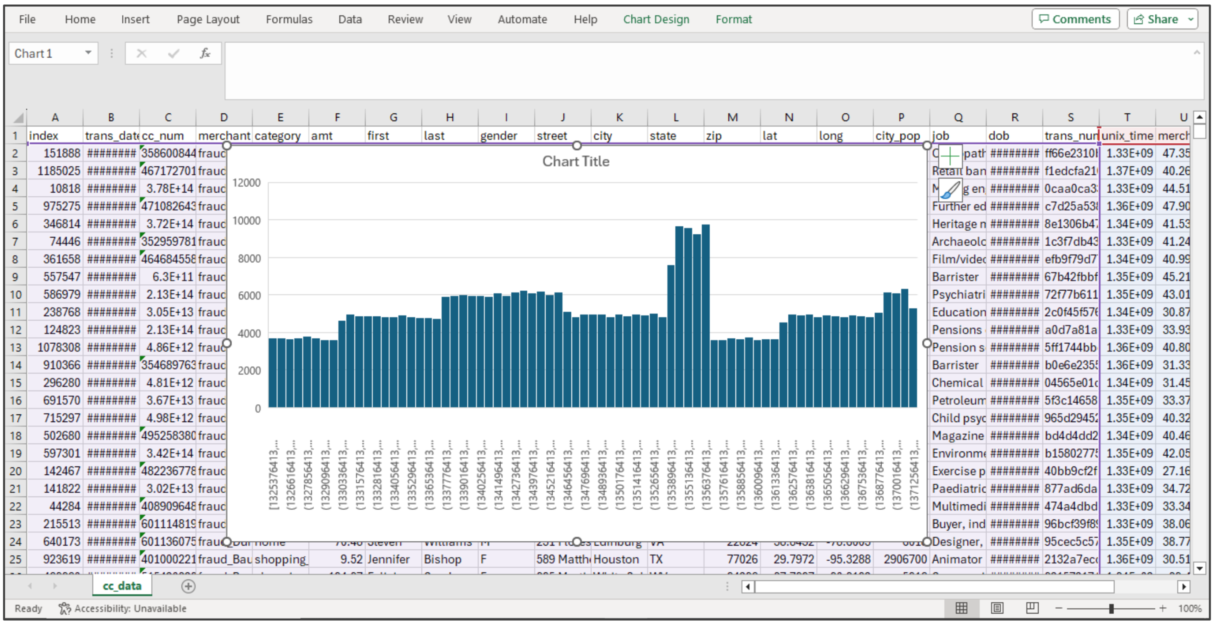


2. Plot histogram on amt

2.1 Select the **amt** column, go to the **Insert** tab, click on either **Insert Statistic Chart**, or choose **Histogram** from the chart options

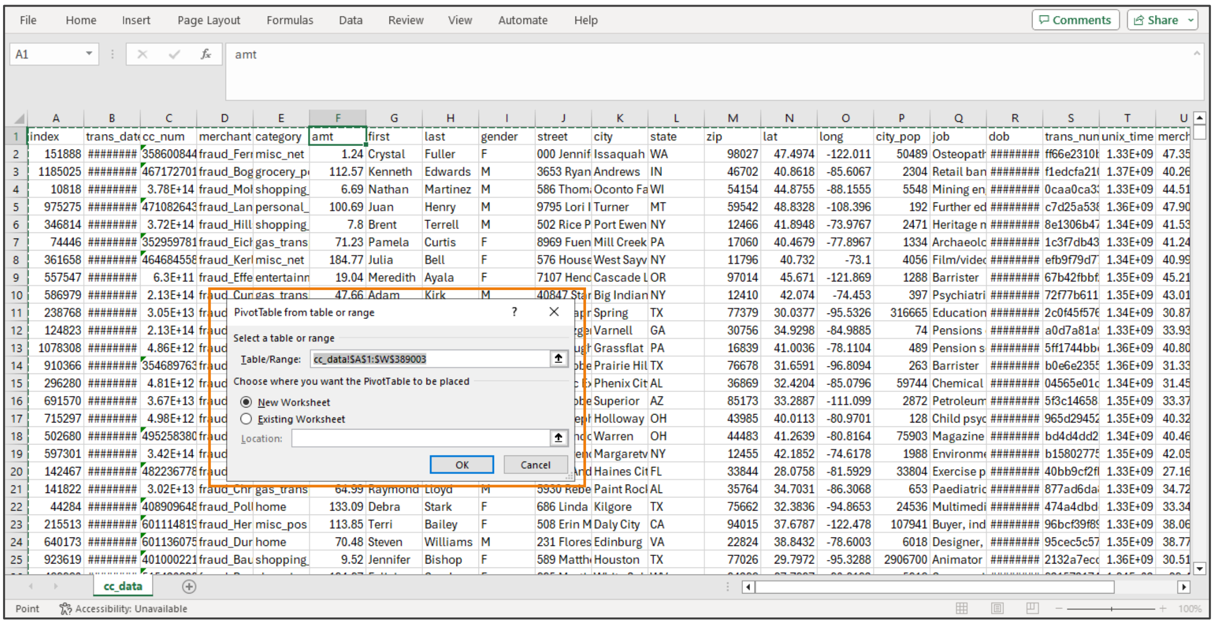


2.2 Adjust the bin size and other settings accordingly

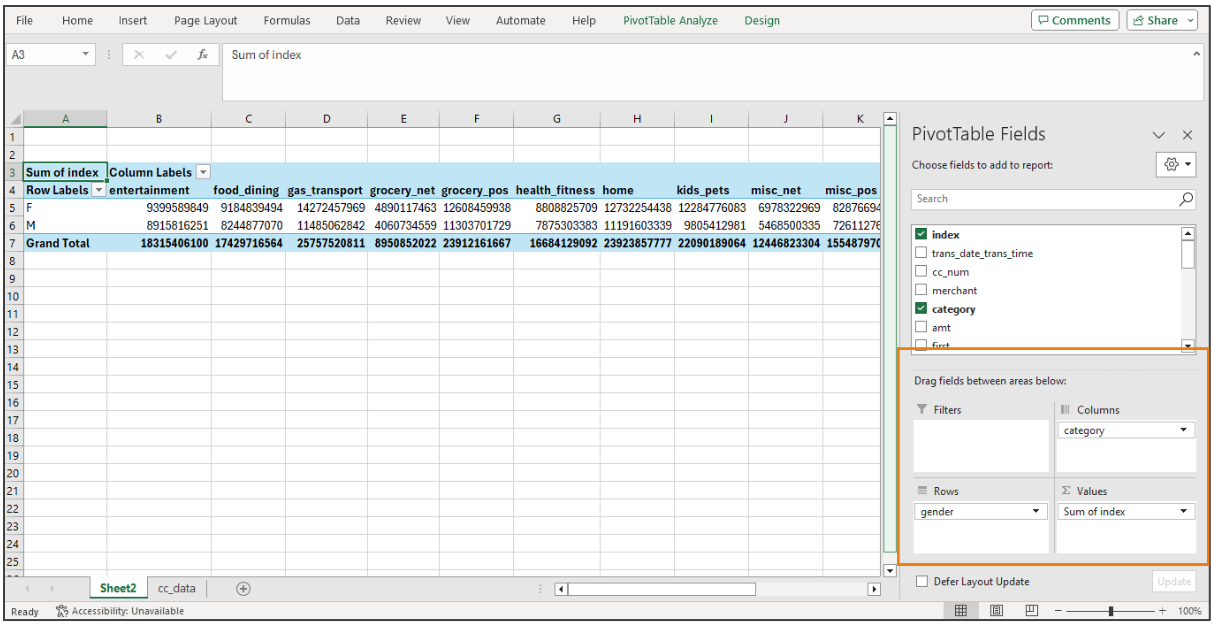


3. Create a report showing number of Frauds by gender and category

3.1 Use a pivot table by selecting the entire dataset, going to the **Insert** tab, clicking on **PivotTable**, and choosing its placement

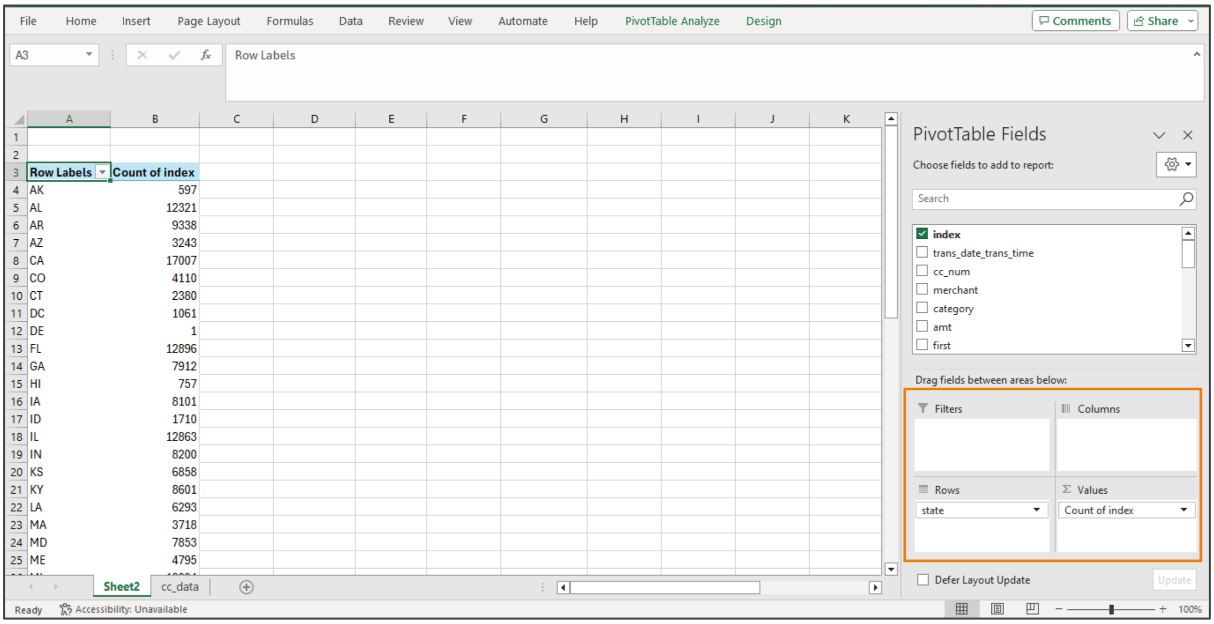


3.2 Drag **Gender** to the rows area, **Category** to the columns area, and the **Count of index** to the values area, then format the pivot table as required

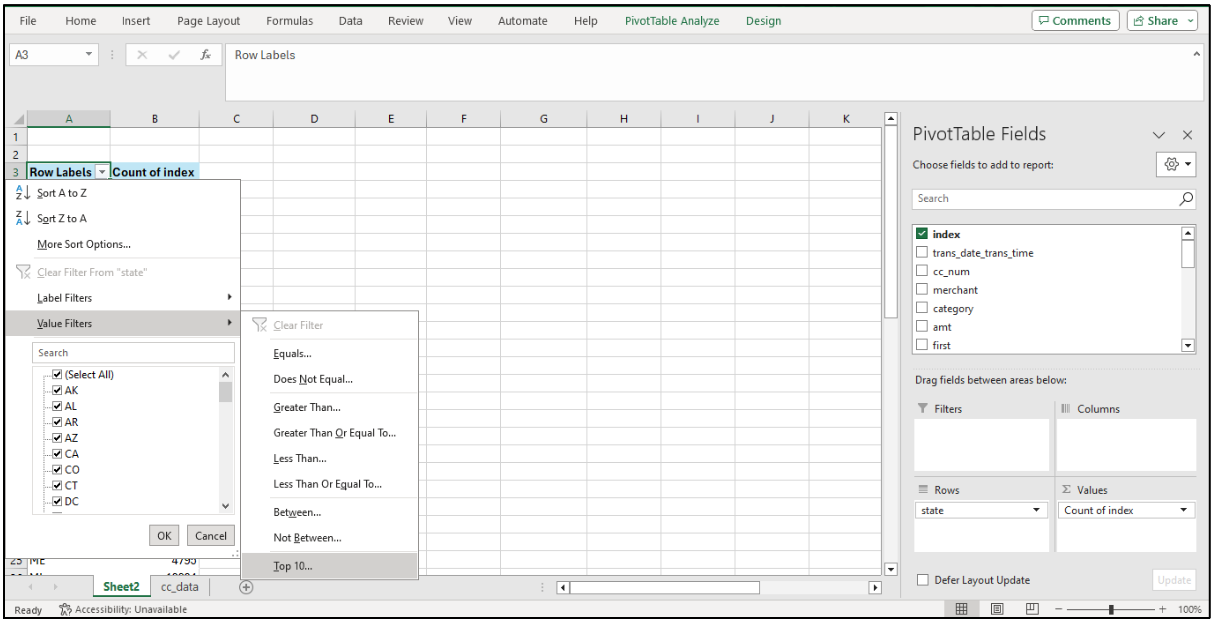


4. Show top 3 states by highest number of transactions

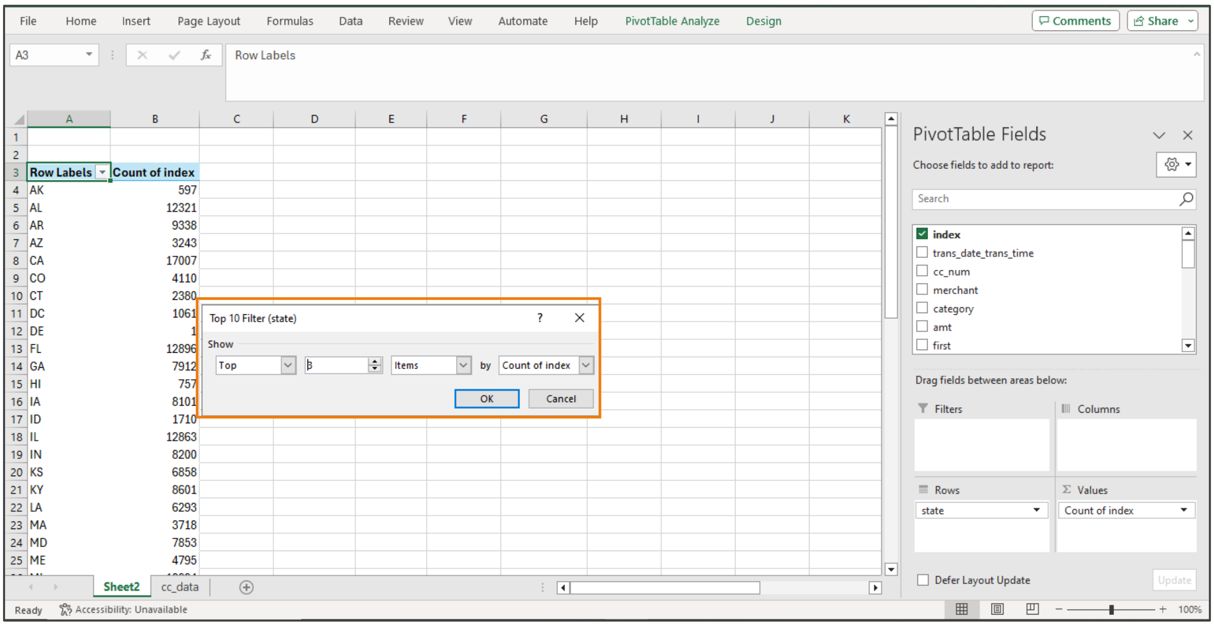
4.1 Use a pivot table by selecting the entire dataset, accessing the **Insert** tab, clicking on **PivotTable** to choose its placement, and then dragging **State** to the rows area and the **Count of index** to the values area



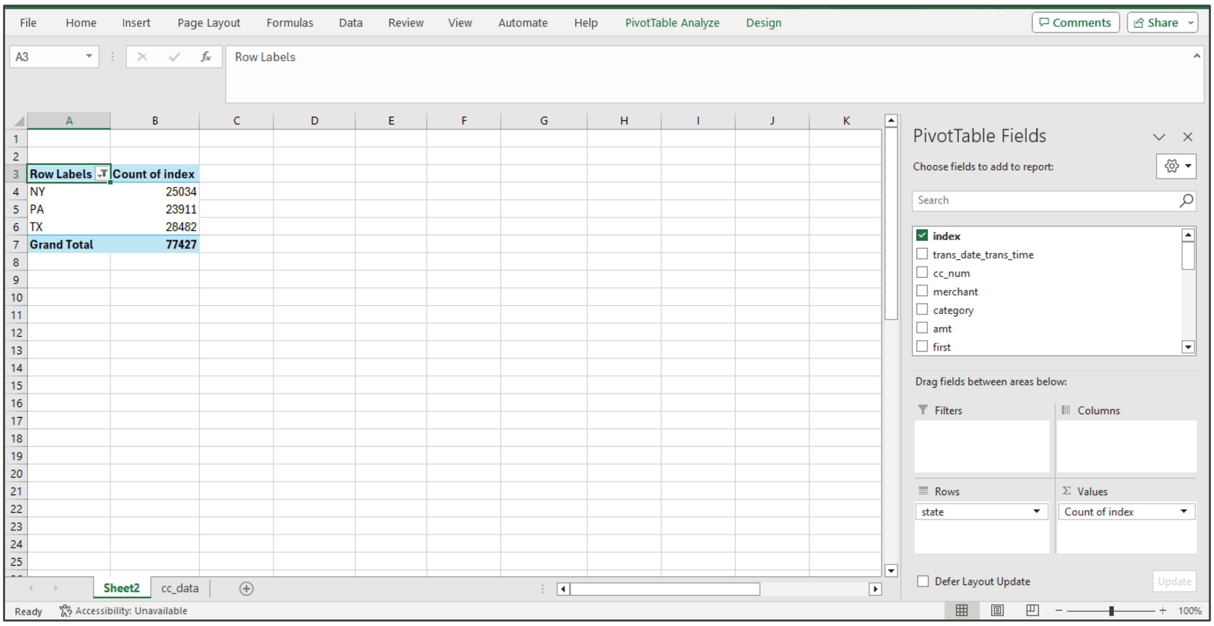
4.2 Filter the pivot table by clicking on Row Labels, selecting **Value Filters**, and then choosing **Top 10**



4.3 Select **Top**, **3**, **Items**, and by as **Count of index** in the **Top 10 Filter (state)** window



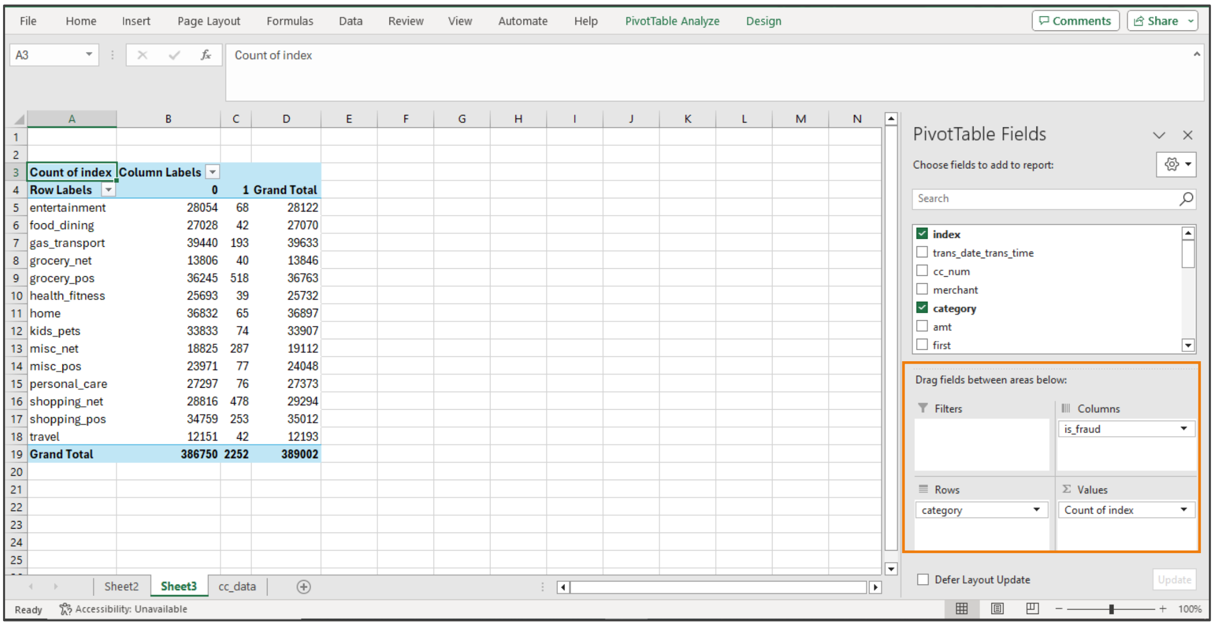
Here are the top 3 states by highest number of transactions.



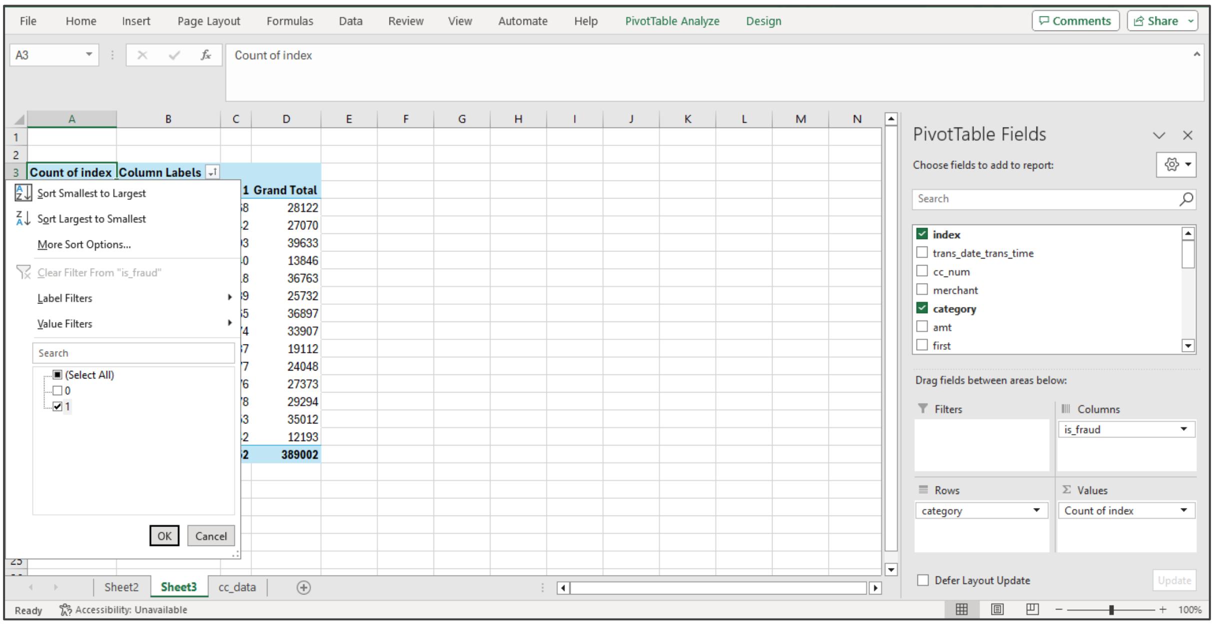
Data Analysis in Excel

1. Is there a correlation between transaction amount and city population?
   Yes

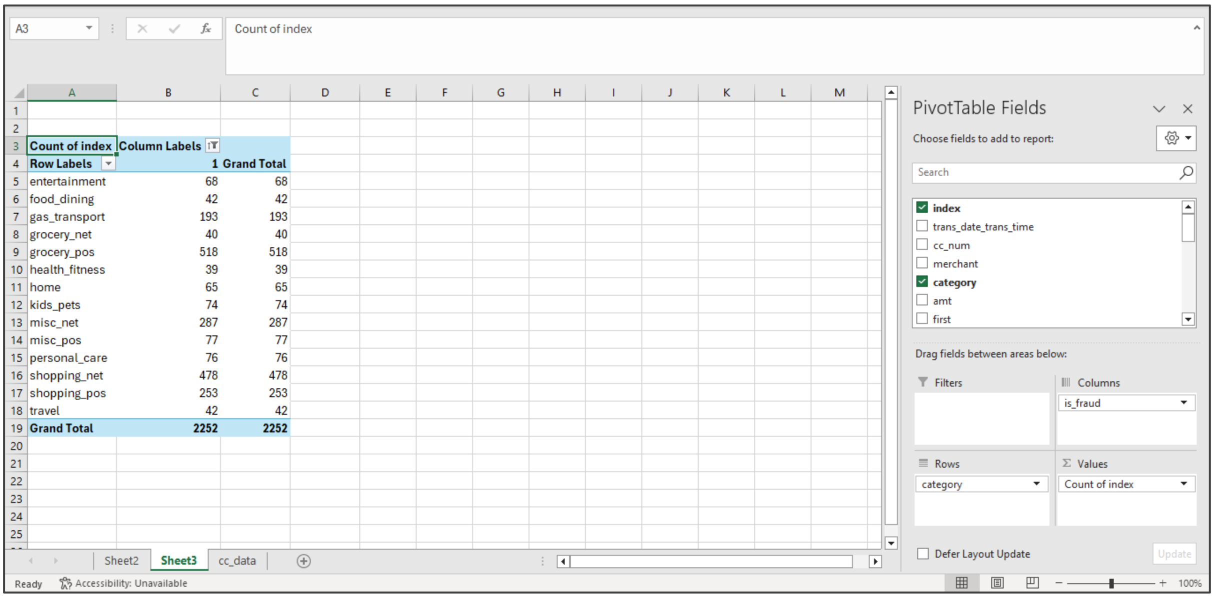
2. How many fraudulent transactions occurred in each category?
   2.1 Create a PivotTable by selecting the entire dataset, including the **Category** and **is_fraud** columns, then go to the **Insert** tab, click on **PivotTable**, choose its placement, and click **OK** to finalize
   2.2 Drag the **Category** field to the rows area, the **is_fraud** field to the column area, and **index** into the value field, selecting count as the aggregate function



2.3 To filter only fraudulent transactions, click on the arrow next to **Column Labels** in the PivotTable, and then check only **1**

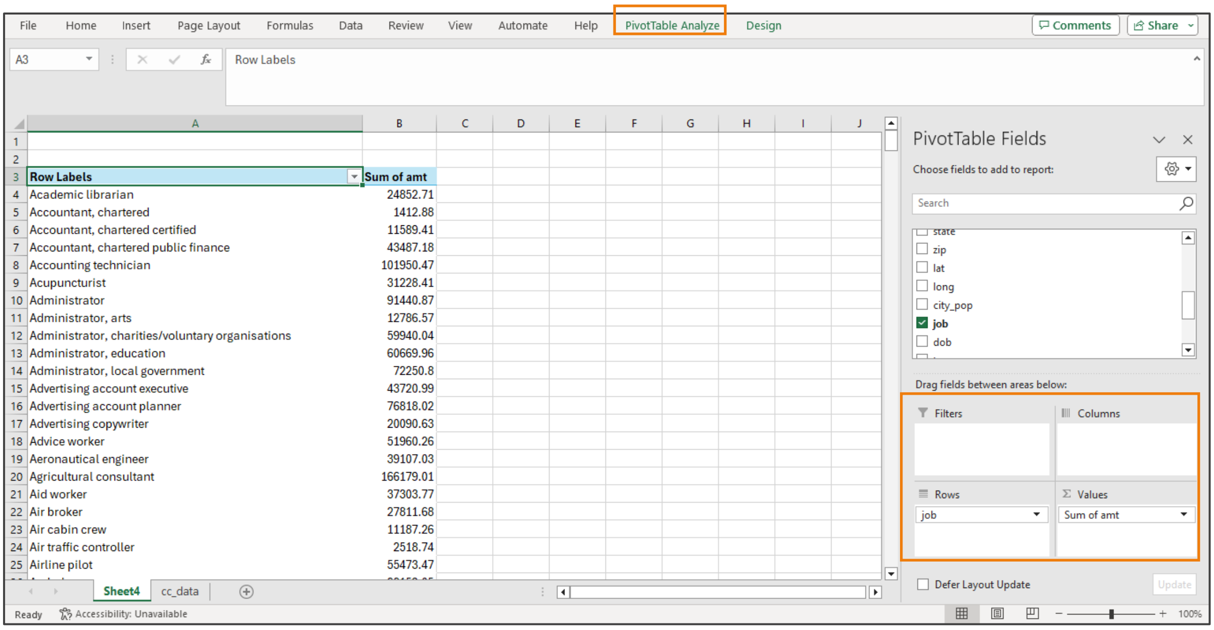


Here, it results in displaying only the count of fraudulent transactions for each category in the pivot table.

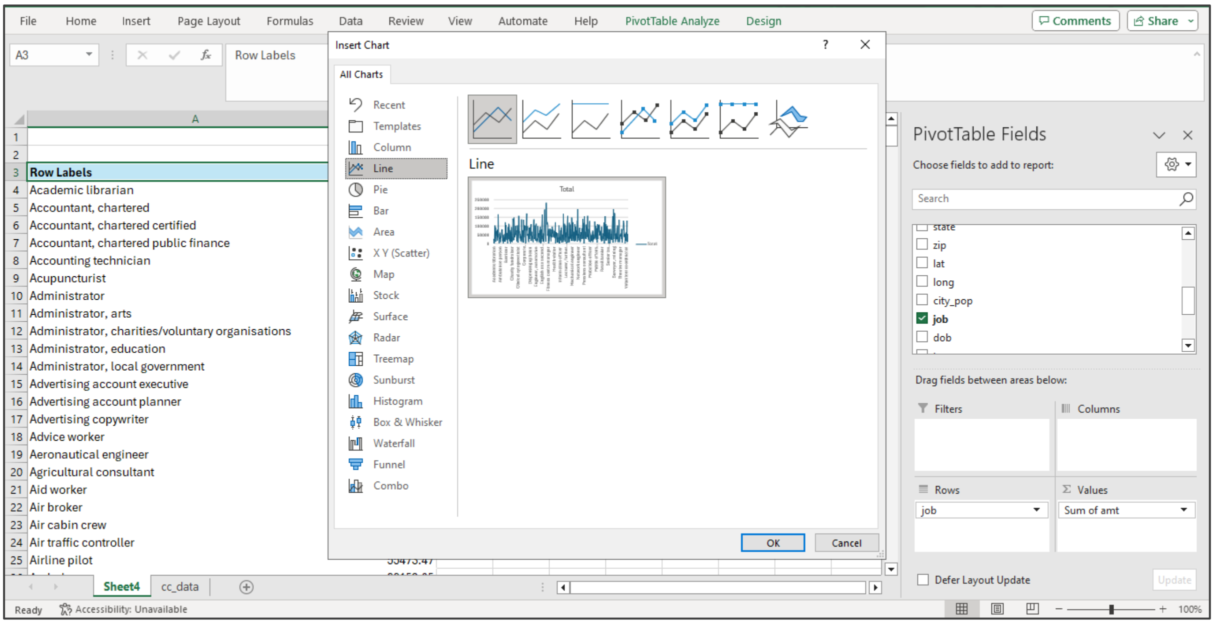


3. How does the average transaction amount vary between different job roles?

   3.1 Create a pivot table, dragging **job** role to the rows area and **amt** to the values area, then format the pivot table as necessary



   3.2 In the pivot table screen, click on **PivotTable Analyze**, choose **Pivot Charts** to create a line chart



Hence the chart is created.